

evaluation boards that developers can use to quickly develop related products. For instance, there are special evaluation boards for use by those building energy meters, sensors, etc.

Launched recently, Analog Devices' ADE7878 smart energy meter evaluation board can be used together with accompanying software to implement a fully functional three-phase energy meter. It comprises an ADE7878 energy meter IC and a microcontroller.

Another example of a specialised product is Optys Corporation's evaluation board for integrated circuits (ICs) of universal frequency-to-digital converter UFDC-1 and its high-speed version UFDC-1M-16. The board is designed for high-performance sensor systems. The user can connect sensor outputs to the evaluation board, measure any output frequency-time parameters and test the sensor system's functions immediately. This saves development cost, design time and the effort of data acquisition and sensor system prototyping.

The development kit includes the evaluation board equipped with a PC interface, an IC of UFDC-1 or UFDC-1M-16 in a 28-lead plastic dual-inline package (PDIP) and the required support software. It can be used for developing digital and smart sensor systems, frequency counters, tachometers and tachometric systems, data acquisition systems, virtual instruments and other applications.

Yet another high-tech example is Mistral's new radar receiver, SaraAmsh. This 1U form-factor, 48cm sub-rack-mountable, compact unit has an 8-channel receiver that supports direct intermediate frequency (IF) sampling. It is built around a Virtex-5 FPGA-based processing engine with powerful processing and high-bandwidth features, making it ideal for DSP applications like signal intelligence, electronic warfare and counter measures.

A. Venkatakrishna, vice president-hardware design, Mistral Solu-

Commonly used hardware tools

1. Development boards
2. Evaluation boards
3. Single-board computers
4. System-on-module
5. Logic analysers
6. Oscilloscopes
7. Spectrum analysers
8. Emulators
9. Field-programmable gate arrays
10. Hardware trace solutions
11. Multimeters

tions, explains, "It integrates multiple analogue-to-digital converters, FPGA processing engines, a single-board computer and various serial and parallel industry standard interfaces, to provide customers a complete solution. Customers can focus on their end application instead of spending valuable time integrating multiple commercial, off-the-shelf boards available in the market to build their own radar receiver solution."

Faster, better, cheaper

Today's development tools not only enable faster creation of highly-capable embedded systems, but also ensure as low a cost as possible. That is one of the key features of Agilent Technologies' Medalist i3070 Series 5 in-circuit test (ICT) platform launched last year.

The platform combines a new analogue measurement technology with the industry's fastest 12MHz hybrid pin card, giving electronics manufacturers 20 to 30 per cent throughput improvement over the current i3070 at the same listed price. This enables contract manufacturers to considerably reduce their cost of test while increasing coverage.

According to the company, the Series 5 ICT system also offers a new infrastructure that allows manufacturers to incorporate external circuits easily for added test coverage while providing better control of these circuits. Instead of having multiple sets of external circuits in multiple fix-

tures, manufacturers can now put the external circuits in the test head. This way, they need only one set per tester, which can save them significant test costs. Agilent has already identified partners to provide high-speed flash programming solutions, LED testing and boundary-scan solutions via these external circuits.

The Series 5 ICT system addresses a broad array of in-circuit and functional test needs, including IEEE1149.6 boundary-scan standard testing and limited-access test applications for highly complex and small-footprint boards in sophisticated consumer electronics, data communication, automotive, aerospace, defence and medical applications.

Meeting the growing demand

It is obvious from the small sample of products covered that hardware tools are essential to improve the quality of embedded systems, as well as speed up the lab-to-market cycle. These tools—ranging from logic analysers and oscilloscopes to FPGA boards and evaluation boards—are in demand for applications ranging from embedded systems education to the development of mobile phones, right up to defence equipment.

Keeping in line with the laws of demand and supply, the industry continues to launch more and more advanced equipment to meet the growing needs of embedded developers. As we sail towards dreams like the Internet of Things and the connected home, the role of embedded systems is obviously going to increase and so also the demand for the tools required for their creation. We will have more to write about when we revisit this topic! ●

Janani Gopalakrishnan Vikram is a freelance writer based in Bengaluru. She writes on a variety of topics, her favourites being technology, cuisine, and life. Chandramouli has a PhD in computer engineering from the University of South Florida, Tampa. He works for a leading EDA company in Bengaluru and writes on a variety of topics in his personal blog